

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures — Senior Data Analyst

Chicago, IL • Sep 2024 – Present

Embedded with Customer Perception (~30 stakeholders), shipping production LLM agents end-to-end

- Designed and shipped a production text-to-SQL copilot over a 67-table BigQuery dataset — RAG over a 12K-query corpus, multi-agent orchestration (Context Researcher → SQL Drafter → Validator → Devil's Advocate), prompt-tuned per stage against 23 golden-query test cases and 8 deterministic SQL guardrails. 95% SQL accuracy, ~80% faster query creation; first AI skill in active business use on Data Ventures.
- Built an autonomous Jira ticket-resolution agent with a 6-gate workflow and rigorous per-gate evaluation: MCP-driven context retrieval (FAISS+BM25 over an 11K-ticket archive), plan-then-execute with 10-retry self-healing, 80%+ SQL reuse via similarity search. Closed 400+ data requests across 8 categories — cut turnaround from 30–60 minutes (up to a full day for complex tickets) to under 10 minutes each, with stakeholders never knowing an agent was answering.
- Built 3 MCP servers (24+ tools, stdio) that let every downstream agent reason over the 43-table BigQuery surface with zero hardcoded schema. Same MCP layer now backs the copilot, Jira agent, autonomous analyst, and hybrid orchestrator.
- Architected a 9-stage gated autonomous analytics pipeline (SELECT → PLAN → EXECUTE → INTERPRET → PACKAGE → CHART → NARRATE → VALIDATE → COMMIT) on a 740K-panelist dataset; 146 published analyses in 6 weeks with analyst review concentrated at the publish gate. Prompts and gates iterated on a golden-set harness reused across every agent in the portfolio.
- Co-built a hybrid orchestrator with Data Science and Product: routes business questions in parallel to a Cube.js semantic-layer agent and a context-engineered agent, reconciles outputs, and turns disagreements into ratified semantic-layer entries — a self-growing definition library shared across analytics agents.
- Productized two AI skills (cp-analytics, Simple CP Interaction) org-wide as official internal Wibey (Walmart's Claude Code fork) skills after Product + Data Science partnership review — first AI skills productized across Walmart Data Ventures. Recognized with a Walmart Making a Difference Award.

FTI Consulting — AI Engineer (UChicago MSADS Capstone)

Chicago, IL • Jan – Aug 2024

Embedded with the client, shipped into their workflow

- Deployed a RAG application for consulting-proposal generation adopted directly into FTI's workflow — query rewriting (+34% generation quality), adaptive chunking (+42% retrieval), Dockerized Streamlit on AWS EC2; cut proposal time ~30%. Best in Show — UChicago MSADS Capstone.

University of Chicago — Graduate Researcher, Data & Democracy Initiative

Chicago, IL • Jan – Sep 2024

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment with controlled prompt/temperature conditions; A/B tests persuaded 15% of participants to donate to an opposing cause.

Baker Tilly — Consultant, Enterprise Platform Management (Oracle)

Chicago, IL • Sep 2021 – Sep 2022

- Delivered Oracle enterprise implementations end-to-end with client business and IT stakeholders — including RPA workflows automating high-volume client operations and AWS computer-vision deployments from problem framing through production.

Innovare (EdTech Startup) — Data Scientist

Chicago, IL • Oct 2022 – Mar 2023

- Logistic-regression student-risk model across 10+ districts (5,000+ students): +7% retention, -15% course failures; automated the supporting ELT into BigQuery (+23% reporting accuracy).

SELECTED PROJECT

- NL-to-SQL Copilot (live demo)** — public mirror of the Walmart architecture: Streamlit + LangChain + Gemini 2.5 + RAG over historical queries + BigQuery execution + auto-visualization, Dockerized on GCP Cloud Run. sql-rag-frontend-simple-481433773942.us-central1.run.app

SKILLS

Agentic / LLM stack: Production LLM agents (deployed, evaluated, iterated on), multi-agent orchestration, MCP (Model Context Protocol) server design, tool development & evaluation, prompt engineering & optimization, context engineering, RAG (embeddings + FAISS + BM25 + RRF), evaluation harnesses (golden-set, gated pipelines, human-in-the-loop), LangChain, ChromaDB

Models: Anthropic Claude (Opus / Sonnet / Haiku), Cohere Command (familiar from public API), Google Gemini (2.5 Pro / Flash, Vertex AI), OpenAI, multi-model routing

Customer & delivery: Customer-embedded delivery (FTI client + Walmart stakeholders), metric definition with non-technical partners (quality / efficiency / accuracy), bridging SWE ↔ ML ↔ business, enterprise rollout & governance review

Engineering / data: Python, SQL, FastAPI, Docker, asyncio, mypy --strict, pytest; BigQuery, Databricks, MySQL, dbt; GCP (Cloud Run, Vertex AI), AWS (EC2); Slack webhooks, cron, Playwright SSO

AI dev environment: Claude Code (daily — via Wibey, Walmart's internal Claude Code fork), authored + productized Claude Code skills, agentic coding workflows

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0) — Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5) — Chicago, IL

HONORS

- Walmart Making a Difference Award — 2026 (for 2025 AI/agent portfolio impact across Customer Perception)
- Best in Show — UChicago MSADS Capstone (FTI RAG proposal generator)